

Iteration 3 Proposal:

1. Project Information

Project Title: Automatic Fall Detection using Wearable Inertial Sensors for Health Monitoring

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GitHub Repository: [EECE-5644-Final-Project-Github](#)

Link to Google Drive: [EECE-5644-Final-Project-Dataset](#)

2. Dataset Summary

Dataset field	Summary
Dataset name	FallAllD: A Comprehensive Dataset of Human Falls and Activities of Daily Living
Dataset source	IEEE DataPort; introduced by Saleh, Abbas, and Le Jeannès in IEEE Sensors Journal
Participants	15 subjects
Activities	79 types: 35 falls and 44 ADLs (activity of daily living)
Devices/locations	3 wearable data loggers per subject: waist, wrist, and neck
# of Samples (Instances)	6,605 trial recordings (1,722 falls and 4,883 ADLs)
Repository files	26,420 .dat files and 4 sensor files (Accelerometer, Gyroscope, Magnetometer, Barometer) per recording
# of Features (attributes)	11 sensor channels (accelerometer (x,y,z), gyroscope (x,y,z), magnetometer (x,y,z) and barometer (pressure, temperature))
Target variable (label)	Binary class: ADL (Activity of daily living) = 0; fall = 1

Dataset Description and Suitability

FallAllD is an open, multivariate time-series dataset that contains simulated falls and normal activities for 15 participants. Each participant wore 3 sensor devices that provided measurements from multiple body positions including wrist, waist and neck. Each recording is represented by 4 files: triaxial accelerometer (A), barometer (B), triaxial gyroscope (G) and triaxial magnetometer (M).

Sensor	Channels	Sampling rate
Accelerometer	x, y, z acceleration (3)	238 Hz
Barometer	Pressure and temperature (2)	10 Hz
Gyroscope	x, y, z angular velocity (3)	238 Hz
Magnetometer	x, y, z magnetic field (3)	80 Hz

The sensor streams have different sequence lengths because they were sampled at different rates. So, they must be aligned, resampled, or summarized during preprocessing. This dataset is appropriate for this project and determining falls because it provides labeled examples of the positive class (35 kinds of falls) and the negative class (44 Activities of Daily Living). This dataset also includes measurements from multiple body positions and participants.

Each individual data file has a different name based on the device, activity, trial and sensor. A filename such as S01_D1_A013_T01_A.dat identifies subject S01, device D1, activity A013, trial T01, and the accelerometer sensor A. The same recording key has corresponding B, G, and M files. Thus, the 26,420 individual files represent 6,605 complete sensor-device recordings.

The supplied ActivityID2Str.m mapping defines IDs 1–44 as activities of daily living and IDs 101–135 as falls. Accordingly, A001–A044 are encoded as ADL (0), while A101–A135 are encoded as fall (1). For example, S01_D1_A013_T01_A.dat is an ADL recording, while S01_D1_A101_T01_A.dat is a fall recording.

3. Data Preprocessing

To start, we converted all the individual .dat files to a singular pkl file containing all the subject/device/activity trials. The raw data contained 26,420 files and we started by parsing each filename to get the respective data out and then grouped the 4 sensor files that had the same recording and consolidated it into a single data frame with one row per recording, for a total of 6,605 rows.

Missing and Duplicate Values

Once the pkl file was loaded, we checked for missing values, both at the row level and within each sensor array. No data was missing and no NaNs were present, so no logic was needed to account for missing values. Then we checked for duplicate values and confirmed that there were exactly 6,605 unique keys (composed of subject, device, activity, and trial number) with no duplicates.

Detecting and Treating Outliers

Next, we examined each sensor channel against the known hardware specks (including $\pm 8g$ for the accelerometer, ± 2000 dps for the gyroscope), which we used to flag invalid values. Invalid values are

classified as readings that are non-finite or outside of the range of what the sensor hardware can actually produce. Also, large accelerometer and gyroscope peaks were not removed solely because they were statistical outliers. Those peaks were intentionally left untouched because they could distinguish a fall from an ADL (activity of daily living). And when checking how many samples were invalid, we found there were 47 invalid Gyroscope samples that we corrected using linear interpolation from neighboring valid samples.

Encoding Categorical Variables

We encoded the target variable directly from the ActivityID. So, values from 1–44 were labeled as ADL (0), and values from 101–135 were labeled as fall (1). Subject identifiers were retained only to have a subject-wise split of train, validation, and test data later on. Device location (neck, wrist and waist) was also mapped from the numeric device code and one-hot encoded.

Feature Scaling and Normalization

Before splitting, we scaled each of the 4 raw sensor channels (accelerometer, gyroscope, magnetometer, and barometer) to zero mean and unit variance. This was done because their ranges differed enormously and would otherwise dominate any model by magnitude and not by signal importance. The scaling parameters were only computed using the training data to avoid data leakage, and then they were applied to the validate and test subjects.

Training, Validation, and Testing Split

The training/validation/test split was done at the subject level rather than by splitting individual recording to make sure that no subject's data appeared across more than one split. Using a random fixed seed, 11 subjects were assigned to training, 2 to validation and 2 to test. Also, the fall/ADL class balance was steady across all three splits with roughly 75% ADL and 25% fall in each.

Feature Engineering and Selection

After the data set was scaled and split, we engineered the features and selected specific ones to fit our modeling needs. For each of the 4 sensor channels, we computed 6 statistics: mean, median, standard deviation, minimum, maximum and root mean square. Then we reduced this feature set using variance thresholding to remove constant and near-constant features, along with correlation filtering to drop one feature from every pair correlated more than 95%. This brought the final feature count to 54, which was used to build X_{train} , X_{val} and X_{test} .

4. Machine Learning Model

The project will be done with evaluation of multiple models, comparing how accurate each model performs. Each of the following supervised learning models will be evaluated with accuracy, precision, and recall metrics.

The first model we plan to use is a Logistic Regression model, to train a faster basic model for a simpler baseline. This model will be used for classification between ADL (0) and fall (1) targets and is mostly helpful for establishing a baseline for the following models. We do not expect this to be the highest performing model. The sigmoid function that turns the weighted scores into probabilities is useful for the classification, and simple to interpret. The standardized statistics that are calculated in the data preprocessing are set up nicely to use a Logistic Regression as a baseline, with one significant hyperparameter to tune, which is the regularization strength C .

A second model will be a Support Vector Machine (SVM), which will likely be more accurate than the baseline Logistic Regression model. We can start to evaluate non-linear correlations between the engineered features, established by using an RBF kernel. The scaling that is done in the preprocessing is useful for an SVM analysis, and the model only requires three hyperparameters. The shape of the decision boundary is defined by the kernel, likely an RBF to capture nonlinear correlations not seen by the Logistic Regression. Other hyperparameters are again, the regularization strength C , and the gamma value.

The third model will be a Random Forest model, changing the classification from boundaries to a decision tree based on the engineered statistical features defined in the preprocessing. The correlation is most likely non-linear, so a wider assortment of decisions in a Random Forest model will likely improve significantly from the baseline. It does not require the scaled features, but it does have a strong feature importance and feature interaction capability. This model has more hyperparameters to tune, including the number of trees, max tree depth, minimum samples, and class weight. Overall, this model will most likely outperform the others, so it is important to tune these parameters correctly to emphasize the nonlinearity of the statistical connections.

Lastly, the fourth model will be an XGBoost, which will build a similar decision tree matrix like the Random Forest, but with some slight differences. The sequential trees will be dominant in predicting feature interactions, which are likely to be strong in predicting the fall data. Also, like the Random Forest, there is no need for the scaled features. The additional error minimization comes with a few more hyperparameters. In addition to those mentioned in the Random Forest, there is a need for a learning rate parameter, subsample to reduce overfitting, and samples per tree, due to the number of features created in preprocessing.

These models are useful for comparison by the variance in approach to modeling the data. The Logistic Regression provides a baseline, which can be improved by a more fluent decision boundary

in a Support Vector Machine. The Random Forest will highlight the nonlinear correlations, which can then be improved by the sequential boosting of the XGBoost model.

5. Research Questions

The project aims to find solutions to the following five research questions.

1. How do different supervised machine learning algorithms perform in the classification of falls vs. ADLs from wearable sensor features? Which has the highest classification accuracy?
2. Which sensing methods give the most beneficial prediction information for accurate classification?
3. How impactful is feature selection across linear and non-linear based classification models?
4. Does device placement information improve fall detection classification accuracy?
5. Are any ML models sufficient to predict fall classification based on only the dataset used in this project?

This project will investigate feature engineering and feature selection across a few different supervised learning models. Many of the questions that arise in this project are tailored towards determining the most accurate model for classifying falls versus ADL.

From the preprocessing, there are 6605 samples that are split into training, validation, and testing subsets. In terms of target balance, there is approximately a 75/25 split across all three of the subsets. All missing values are handled by the methods described in the preprocessing section. The feature correlation will be determined by each of the models after training. The preprocessing step has cleaned the data and will be able to be used for model training for the four models described above, to answer the research questions proposed with as much accuracy as possible.